Computer Architecture
(EE 371 / CE 321 / CS 330)

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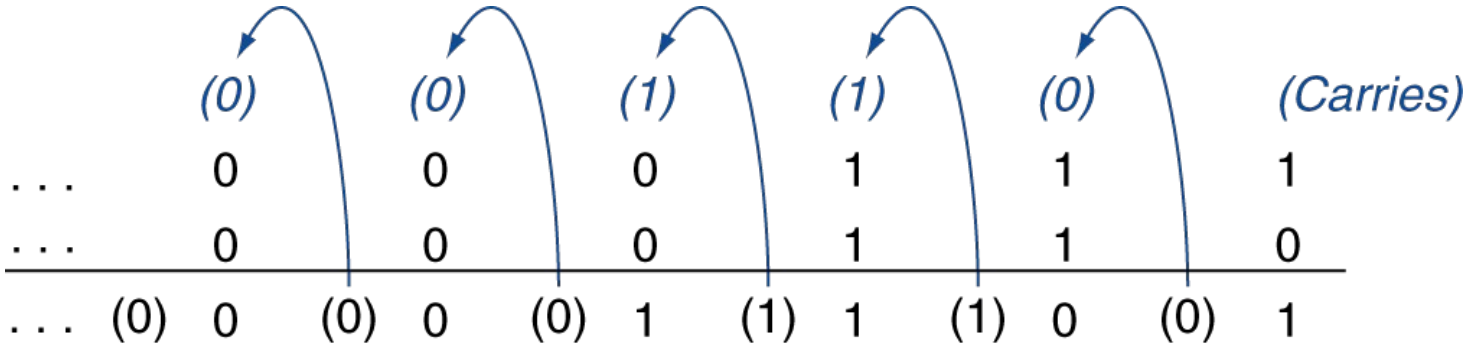
Outline: Unit 3

- Arithmetic for Computers: Review
- Digital Logic Design: Review
- Arithmetic Logic Unit (ALU)
- Single-Cycle Processor Implementation

Integer Addition

$$
\begin{array}{ll}
 & 00000000\ 00000000\ 00000000\ 00000000\ 00000000\ 00000000\ 00000000\ 00000111_{two} = 7_{ten} \\
+ & 00000000\ 00000000\ 00000000\ 00000000\ 00000000\ 00000000\ 00000000\ 00000110_{two} = 6_{ten} \\
\hline
= & 00000000\ 00000000\ 00000000\ 00000000\ 00000000\ 00000000\ 00000000\ 00001101_{two} = 13_{ten}
\end{array}
$$



Integer Addition: Overflow

$$
\begin{array}{rl}
 & 00000000\ 00000000\ 00000000\ 00000000\ 00000000\ 00000000\ 00000000\ 00000111_{two} - 7_{ten} \\
+ & 00000000\ 00000000\ 00000000\ 00000000\ 00000000\ 00000000\ 00000000\ 00000110_{two} - 6_{ten} \\
\hline
- & 00000000\ 00000000\ 00000000\ 00000000\ 00000000\ 00000000\ 00000000\ 00001101_{two} - 13_{ten}
\end{array}
$$

Overflow occurs when the result from an operation cannot be represented with the available hardware, in this case a 64-bit word

- Adding +ve and –ve operands
  - Overflow is not possible
- Adding two +ve number
  - Overflow has occurred if the sign bit of the result is 1
- Adding two –ve numbers
  - Overflow has occurred if the sign bit of the result is 0

Integer Subtraction

- Subtraction through Addition
  - Add first operand and negation of the second operand
  - Example: 7 – 6 = 7 + (-6)

$$
\begin{array}{ll}
 & 00000000\ 00000000\ 00000000\ 00000000\ 00000000\ 00000000\ 00000000\ 00000111_{two} = 7_{ten} \\
+ & 11111111\ 11111111\ 11111111\ 11111111\ 11111111\ 11111111\ 11111111\ 11111010_{two} = -6_{ten} \\
\hline
= & 00000000\ 00000000\ 00000000\ 00000000\ 00000000\ 00000000\ 00000000\ 00000001_{two} = 1_{ten}
\end{array}
$$

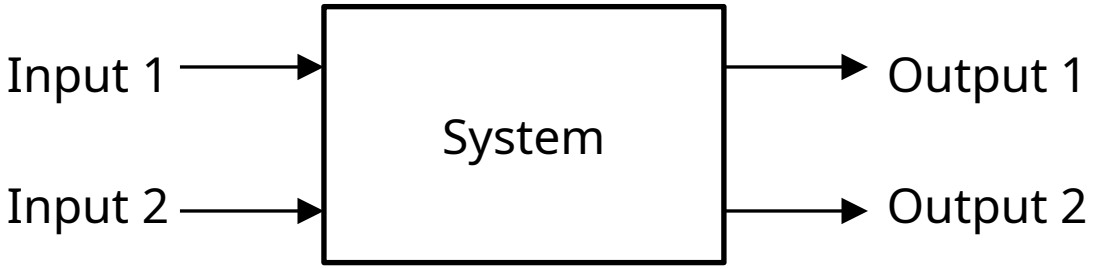
Integer Subtraction: Overflow

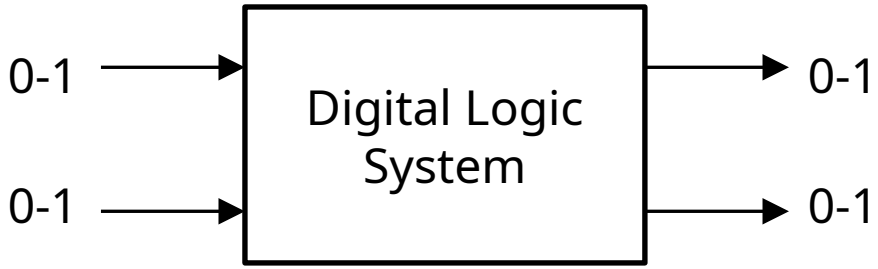
$$
\begin{array}{ll}
 & 00000000\ 00000000\ 00000000\ 00000000\ 00000000\ 00000000\ 00000000\ 00000111_{two} - 7_{ten} \\
+ & 11111111\ 11111111\ 11111111\ 11111111\ 11111111\ 11111111\ 11111111\ 11111010_{two} - -6_{ten} \\
\hline
= & 00000000\ 00000000\ 00000000\ 00000000\ 00000000\ 00000000\ 00000000\ 00000001_{two} - 1_{ten}
\end{array}
$$

Overflow occurs when the result from an operation cannot be represented with the available hardware, in this case a 64-bit word

- Subtracting two +ve or two –ve operands
  - Overflow is not possible
- Subtracting +ve from -ve operand
  - Overflow has occurred if the sign bit of the result is 0
- Subtracting –ve from +ve operands
  - Overflow has occurred if the sign bit of the result is 1

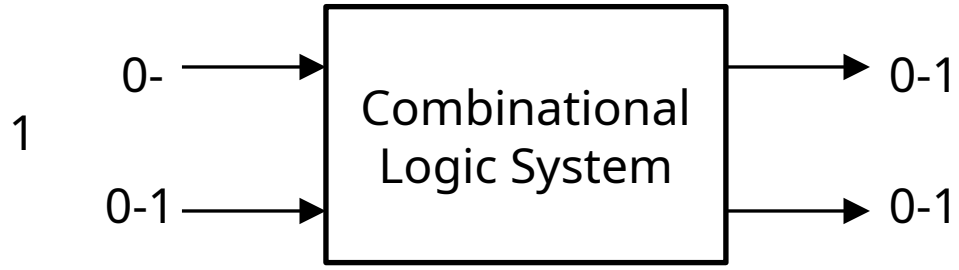
Review of Logic Design



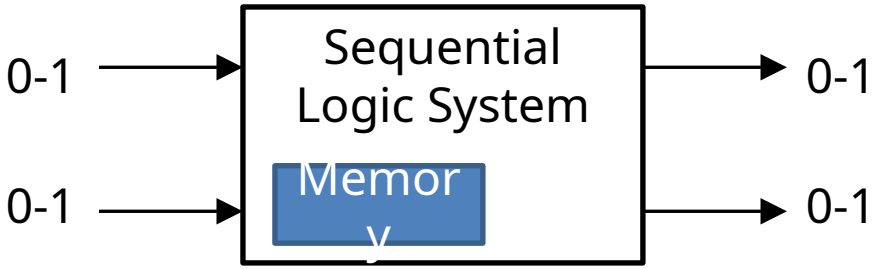


Review of Logic Design

- 2 Types of Logic Systems



The output of a combinational logic system depends only on the current input.



The output of a sequential logic system can depend both on the inputs and internal memory.

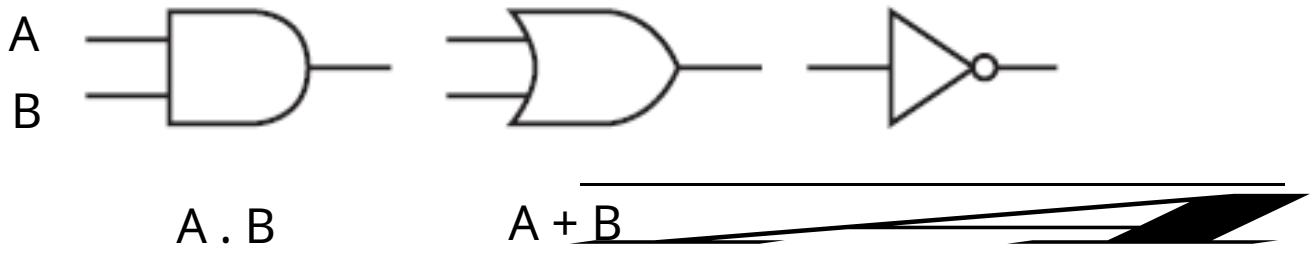
Review of Logic Design

- Truth Table
  - Because a combinational logic block contains no memory, it can be completely specified by defining the values of the outputs for each possible set of input values.
  - Such a description is normally given as a truth table.

Inputs			Outputs		
A	B	C	D	E	F
0	0	0	0	0	0
0	0	1	1	0	0
0	1	0	1	0	0
0	1	1	1	1	0
1	0	0	1	0	0
1	0	1	1	1	0
1	1	0	1	1	0
1	1	1	1	0	1

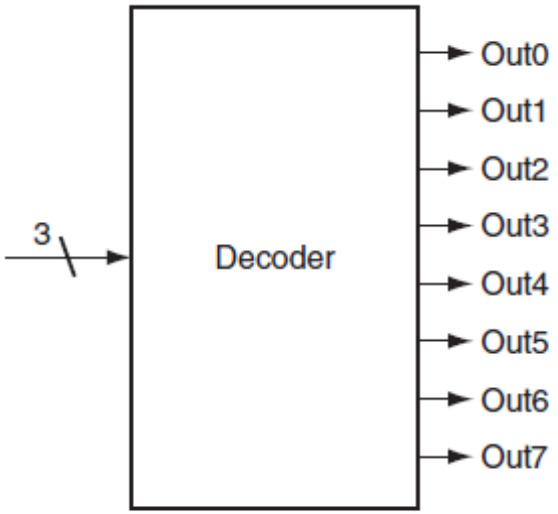
Review of Logic Design

- Logic Gates
  - Logic blocks/systems are built from Logic Gates that implement basic logic functions



Review of Logic Design

- Decoder
  - The decoder has an n-bit input and 2^n outputs, where only one output is asserted for each input combination



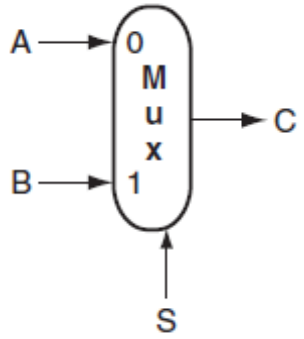
a. A 3-bit decoder

Inputs			Outputs							
12	11	10	Out7	Out6	Out5	Out4	Out3	Out2	Out1	Out0
0	0	0	0	0	0	0	0	0	0	1
0	0	1	0	0	0	0	0	0	1	0
0	1	0	0	0	0	0	0	1	0	0
0	1	1	0	0	0	0	1	0	0	0
1	0	0	0	0	0	1	0	0	0	0
1	0	1	0	0	1	0	0	0	0	0
1	1	0	0	1	0	0	0	0	0	0
1	1	1	1	0	0	0	0	0	0	0

b. The truth table for a 3-bit decoder

Review of Logic Design

- Multiplexor (Selector)
  - Consider the two-input multiplexor.
  - This multiplexor has three inputs: two data values and a selector (or control) value.
  - The selector value determines which of the inputs becomes the output.



Review of Logic Design

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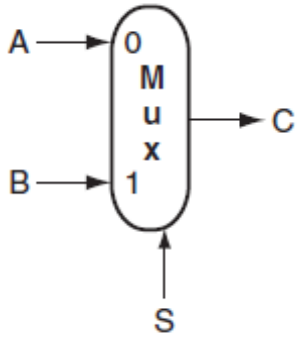
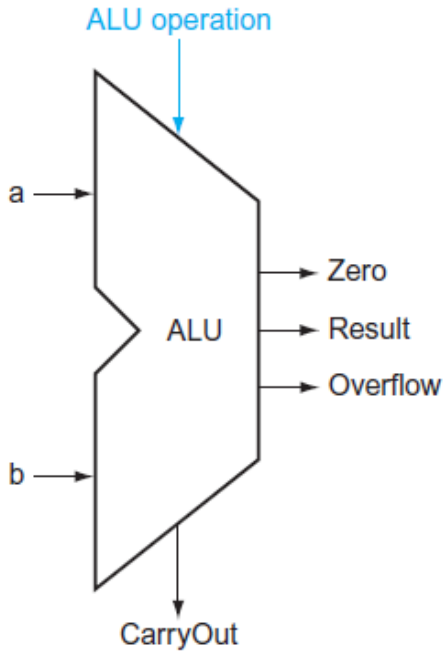


  - Multiplexors can be created with an arbitrary number of data inputs.

Arithmetic Logic Unit (ALU)

- The arithmetic logic unit (ALU) is the brain of the computer, the device that performs the arithmetic operations like addition and subtraction or logical operations like AND and OR.



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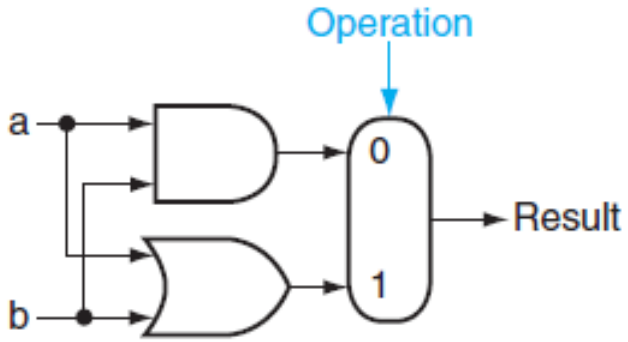
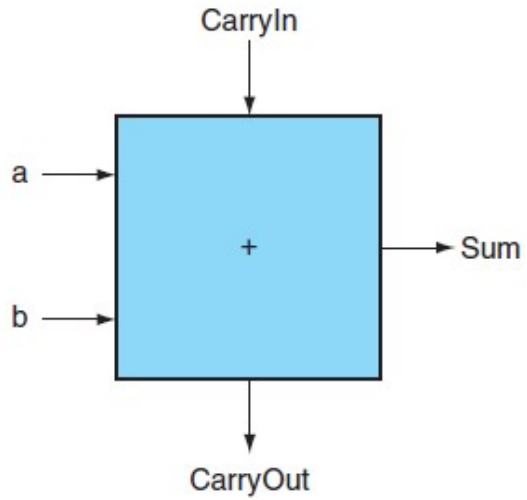
- Let’s focus on 1-bit ALU and progressively build an ALU with more functionality

Arithmetic Logic Unit (ALU)



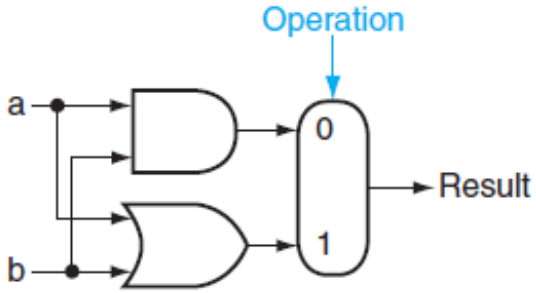
The 1-bit Logical Unit for AND and OR.

1-bit Adder



Inputs			Outputs		Comments
a	b	CarryIn	CarryOut	Sum	
0	0	0	0	0	$0 + 0 + 0 = 00_{two}$
0	0	1	0	1	$0 + 0 + 1 = 01_{two}$
0	1	0	0	1	$0 + 1 + 0 = 01_{two}$
0	1	1	1	0	$0 + 1 + 1 = 10_{two}$
1	0	0	0	1	$1 + 0 + 0 = 01_{two}$
1	0	1	1	0	$1 + 0 + 1 = 10_{two}$
1	1	0	1	0	$1 + 1 + 0 = 10_{two}$
1	1	1	1	1	$1 + 1 + 1 = 11_{two}$

Arithmetic Logic Unit (ALU)



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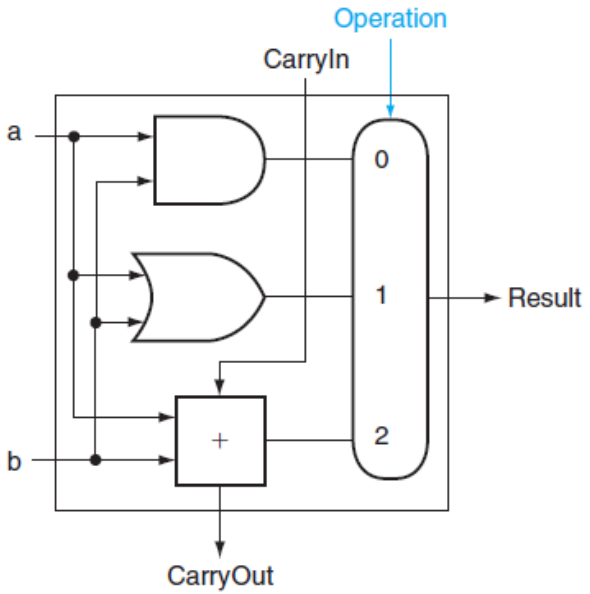
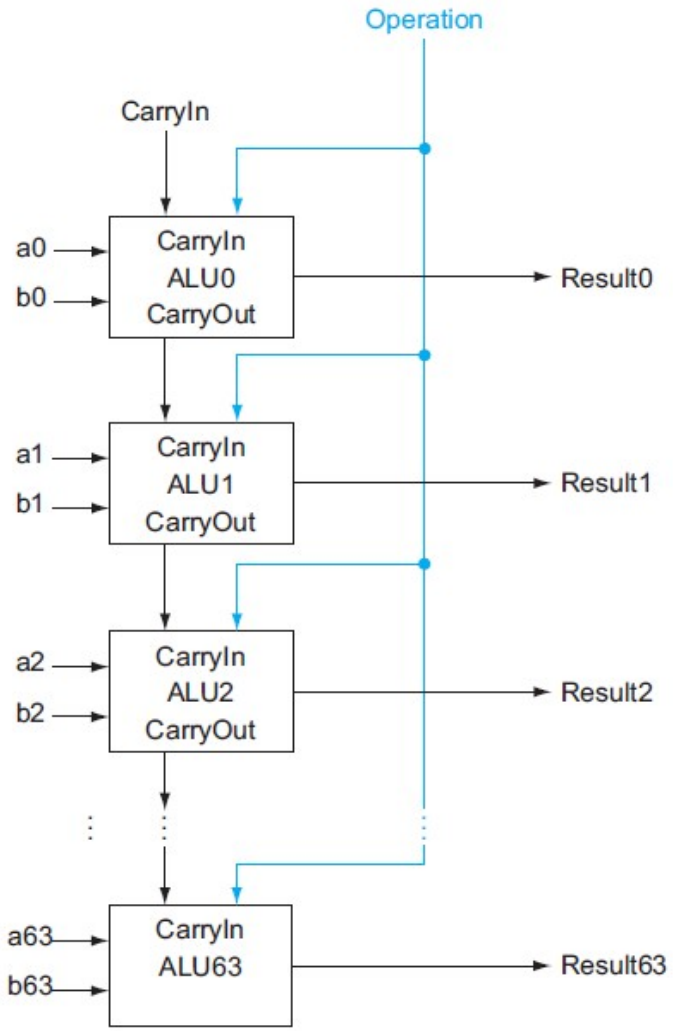


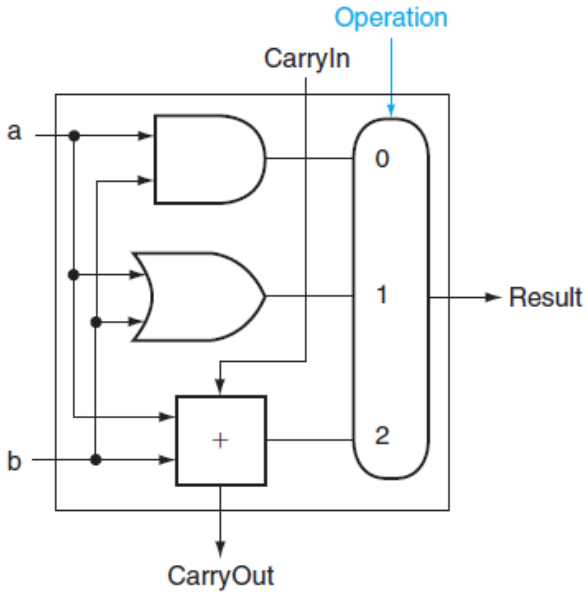
The 1-bit ALU that performs AND, OR, and addition.

Arithmetic Logic Unit (ALU)



A 64-bit ALU constructed from 64 1-bit ALUs

Arithmetic Logic Unit (ALU)



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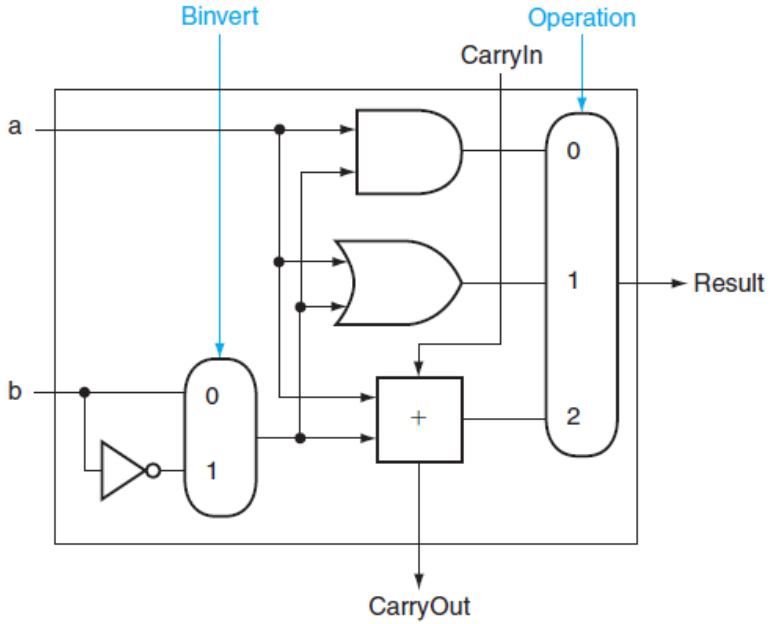
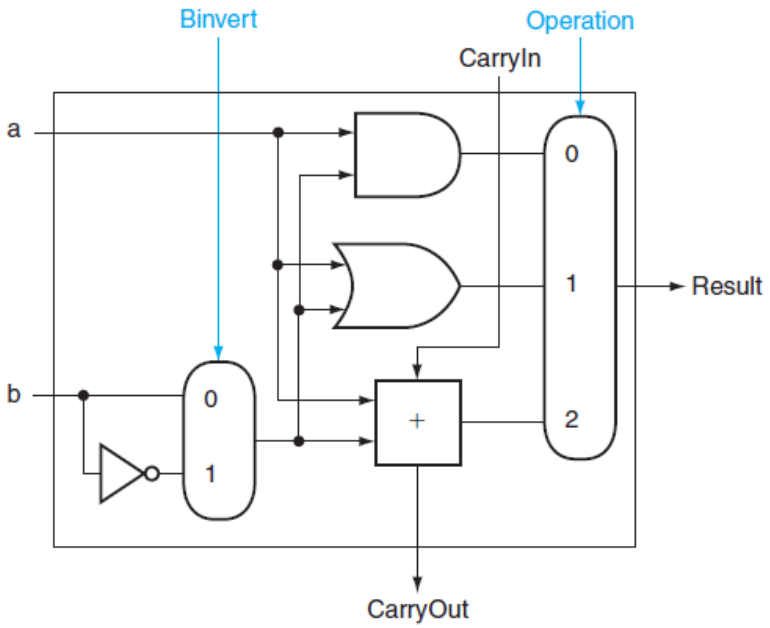
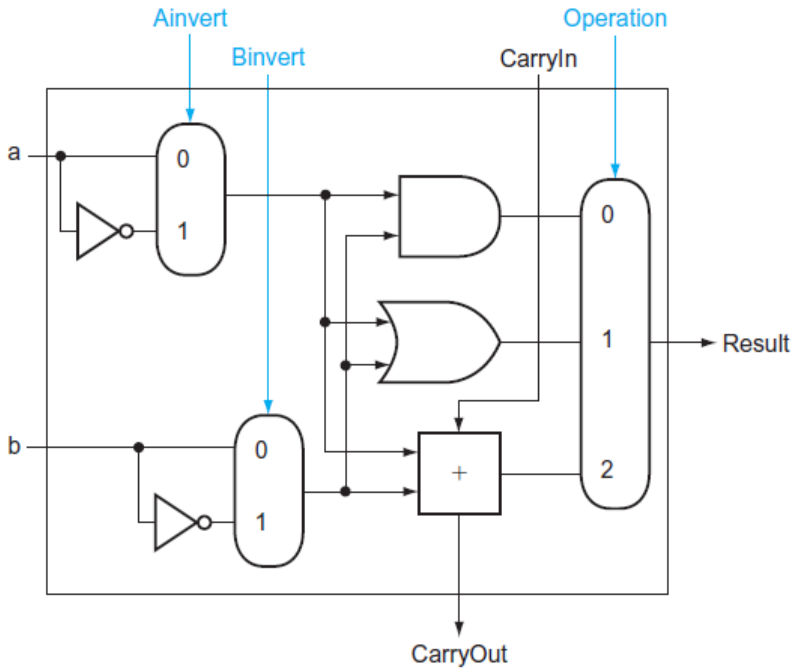


Support for subtraction (through addition).

- By selecting b (Binvert = 1) and setting CarryIn to 1 in the least significant bit of the ALU, we get two's complement subtraction of b from a instead of addition of b to a.

Arithmetic Logic Unit (ALU)





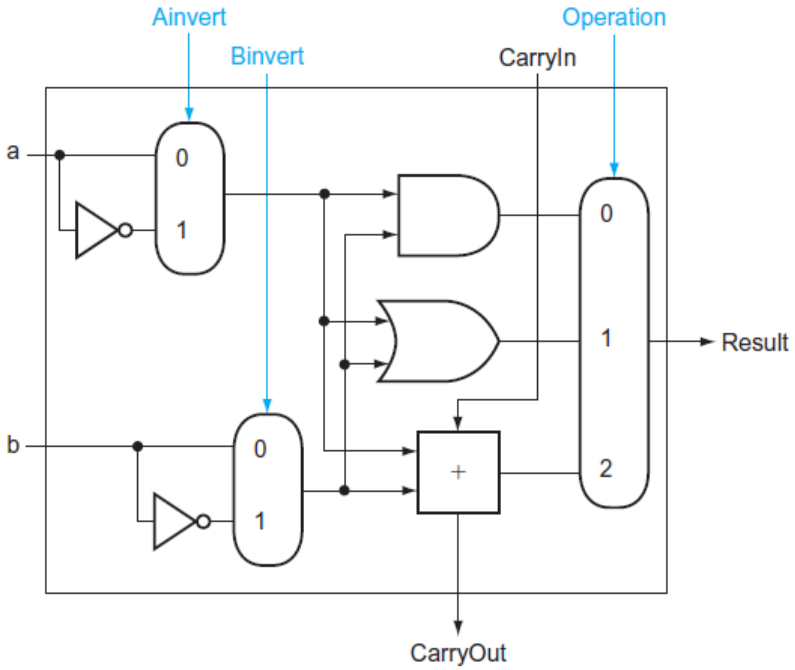
The 1-bit ALU that performs AND, OR, addition, and subtraction (through addition).

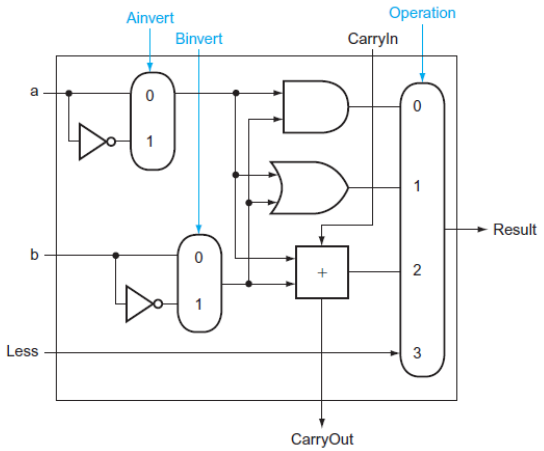
Support for NOR operation (through AND).

- DeMorgan's Theorem .

$$\overline{(a + b)} = \overline{a} \cdot \overline{b}$$

Arithmetic Logic Unit (AL



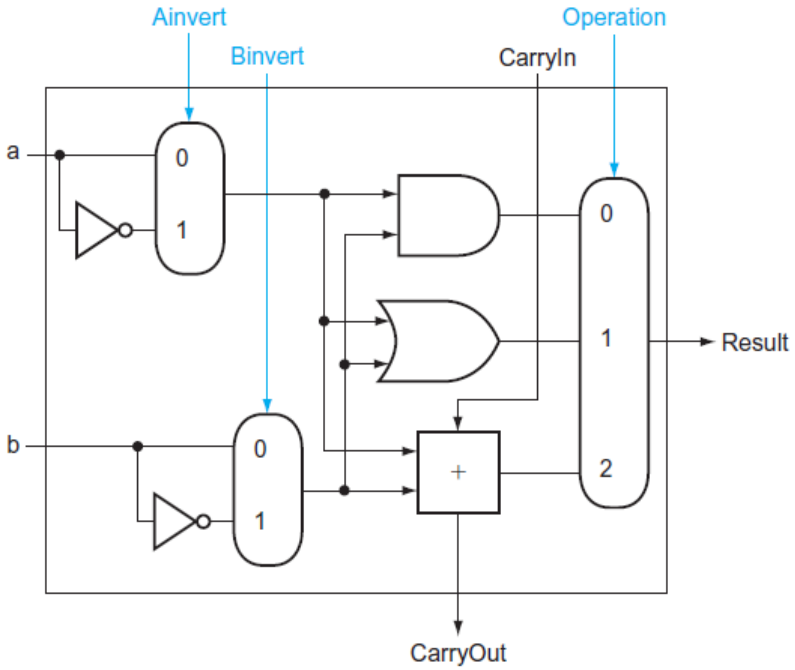


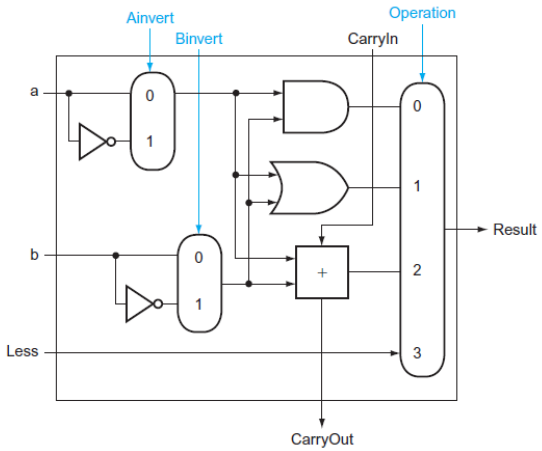
The 1-bit ALU that performs AND, OR, addition, subtraction (through addition) and NOR operation.

Support for slt operation.

- slt rd, rs1, rs2
  rd =1 if rs1<rs2
  rd = 0 otherwise
- Consequently, slt will set all but the least significant bit to 0, with the least significant bit set according to the comparison.

Arithmetic Logic Unit (AL



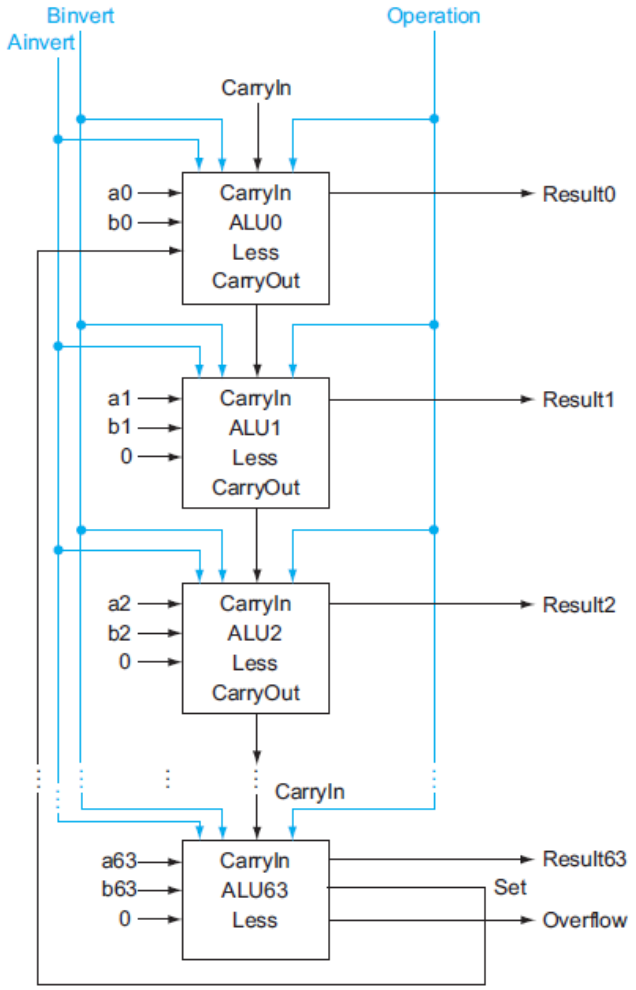


The 1-bit ALU that performs AND, OR, addition, subtraction (through addition) and NOR operation.

Support for slt operation.

- For the ALU to perform slt, we first need to expand the three-input multiplexor to add an input for the slt result. We call that new input Less and use it only for

Arithmetic Logic Unit (ALU)



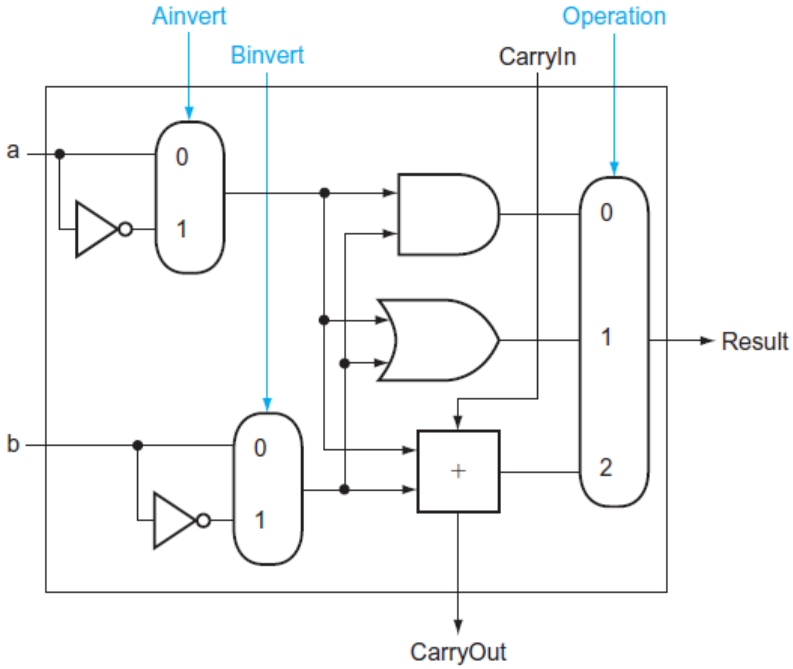
<u>Support for slt operation.</u>

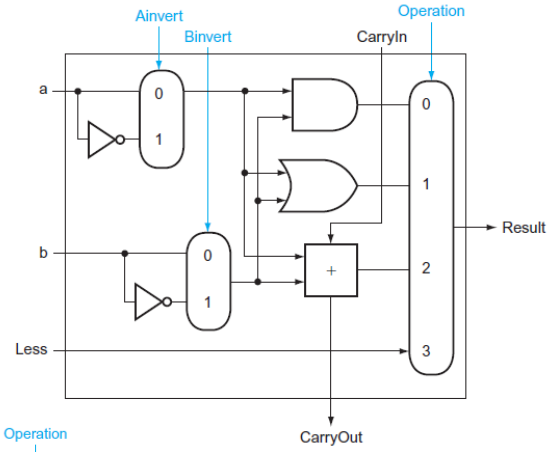
- We must connect 0 to the Less input for the upper 63 bits of the ALU, since those bits are always set to 0.

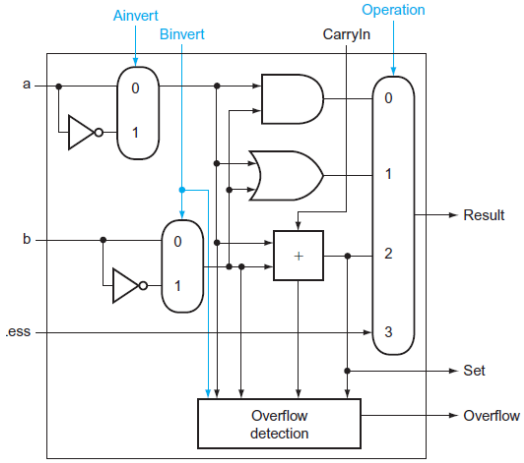
- How to set the least significant bit?
  - We want the least significant bit of slt operation to be a 1 if a < b
  - That is, we want the least significant bit of slt operation to be a 1 if a − b is negative and a 0 if a-b is positive
  - This desired result corresponds exactly to the sign bit values after performing a-b: 1 means negative and 0 means positive
  - So, we need only connect the sign bit (adder output of the most significant bit) to the Less input of least significant bit to get slt.
  - This needs a modified 1-bit ALU at the most significant bit

<u>A 64-bit ALU constructed from 64 1-bit ALUs (Notice the ALU for Most Significant Bit)</u>

Arithmetic Logic Unit (AL







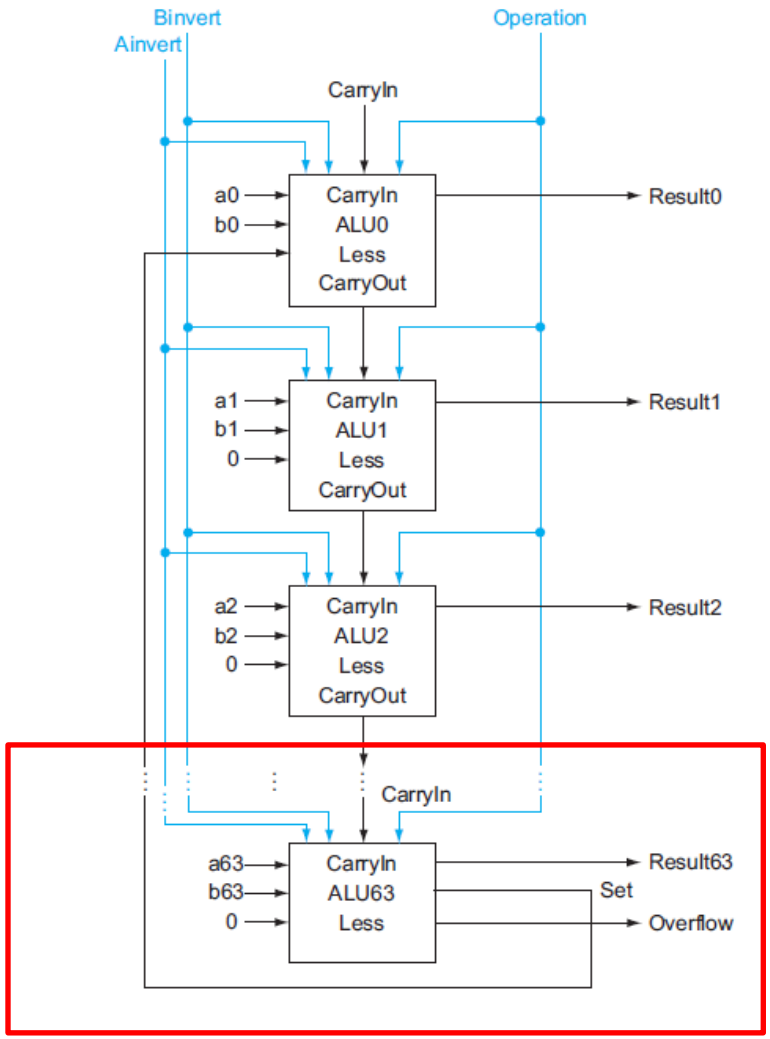
Most significant bit

The 1-bit ALU that performs AND, OR, addition, subtraction (through addition) and NOR operation.

Support for slt operation.

- We need to connect the sign bit (adder output of the most significant bit) to the Less input of least significant bit to get slt.
- This needs a modified 1-bit ALU at the most significant bit (addition of the Set output)

Arithmetic Logic Unit (ALU)



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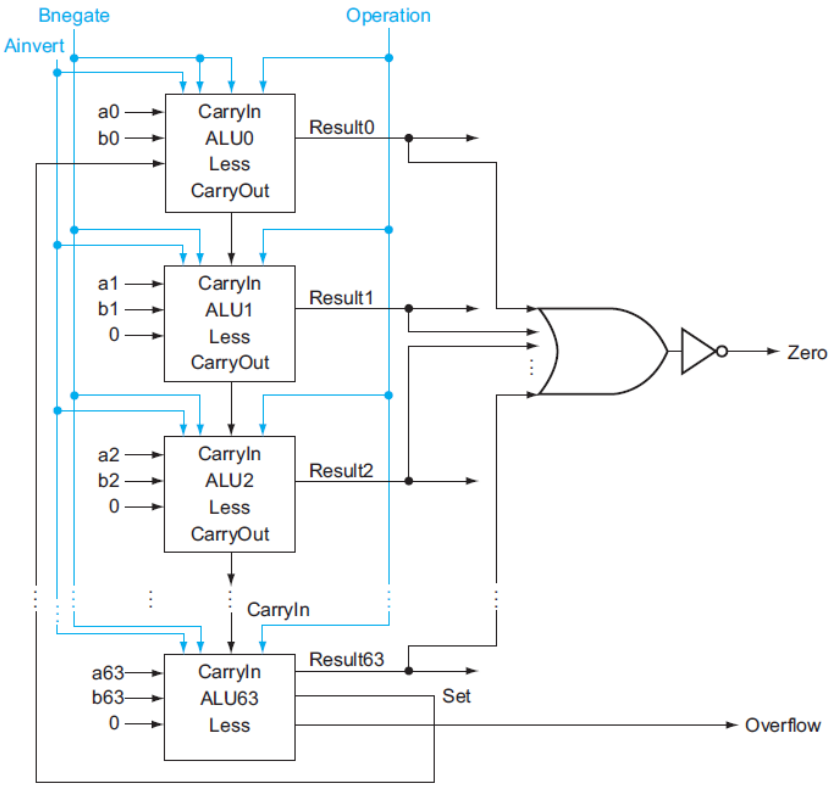
Arithmetic Logic Unit (ALU)

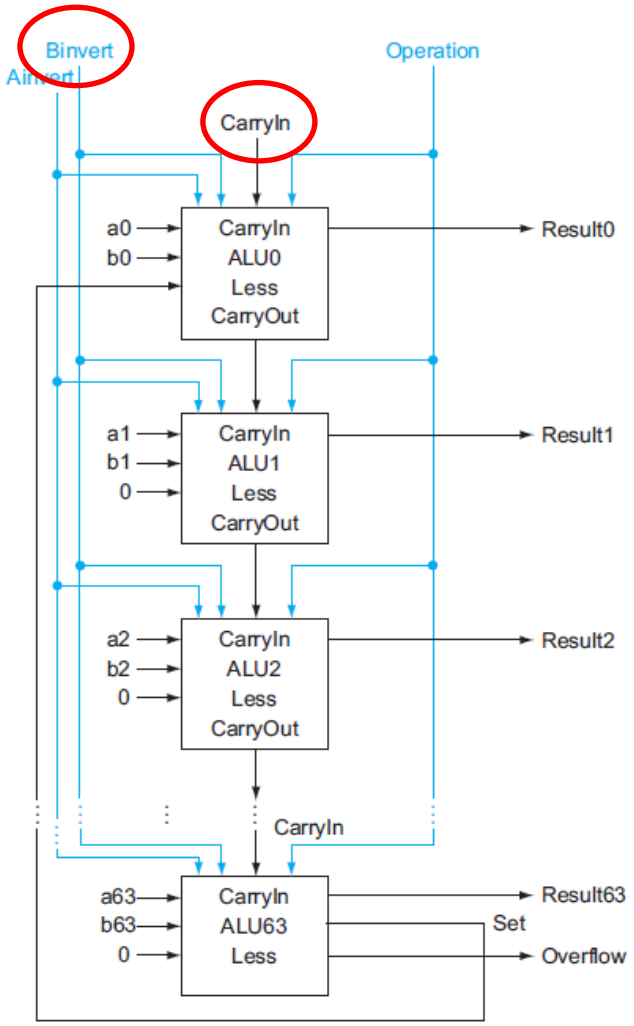


Support for beq operation.

- beq operation branches if two registers are equal.
- The easiest way to test equality with the ALU is to subtract b from a and then test to see if the result is 0.
- To test if the result is 0, the simplest way is to OR all the outputs together and then send that signal through an inverter

$$\text{Zero} = \overline{(\text{Result63} + \text{Result62} + \cdots + \text{Result2} + \text{Result1} + \text{Result0})}$$

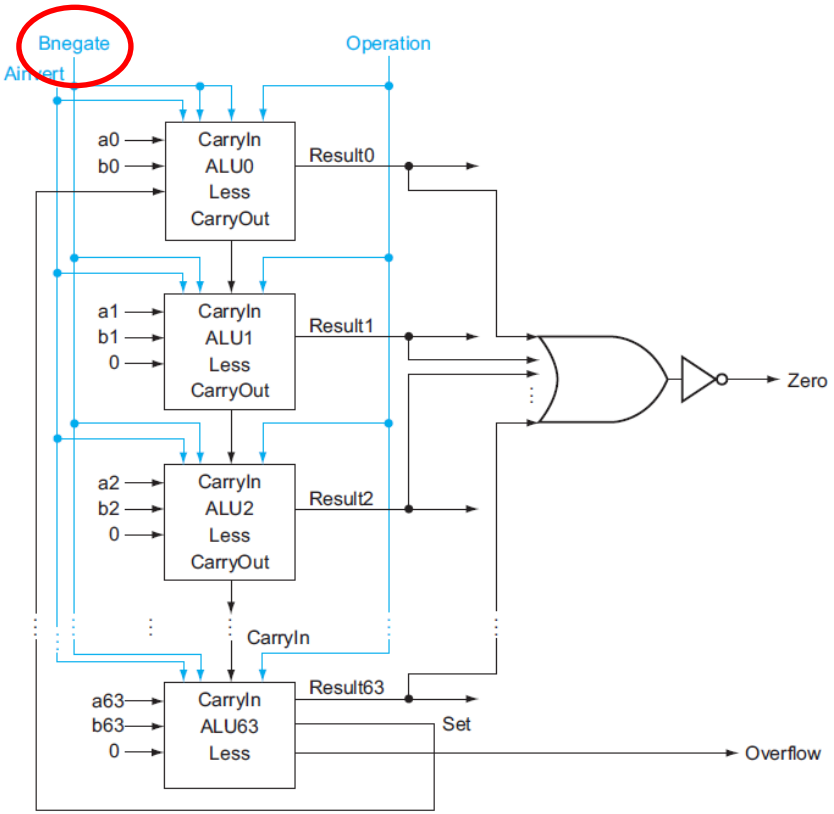
Arithmetic Logic Unit (ALU)



Simplifying Control

- Notice that every time we want the ALU to subtract, we set both CarryIn and Binvert to 1.
- For adds or logical operations, we want both control lines to be 0.
- We can therefore simplify control of the ALU by combining the CarryIn and Binvert to a single control line called Bnegate.

Arithmetic Logic Unit (ALU)



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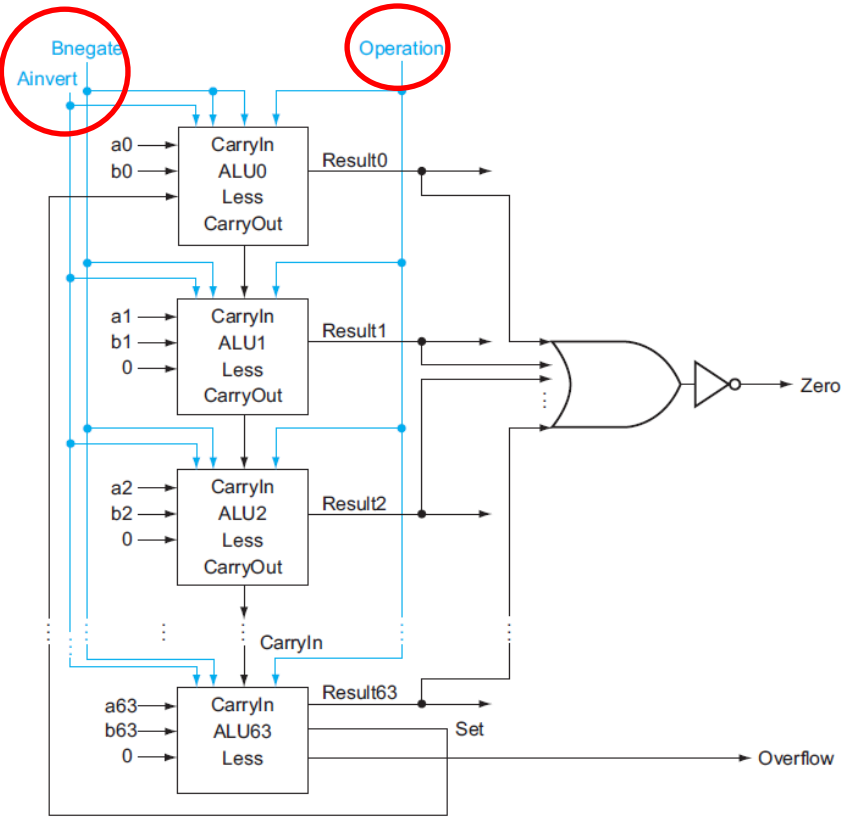
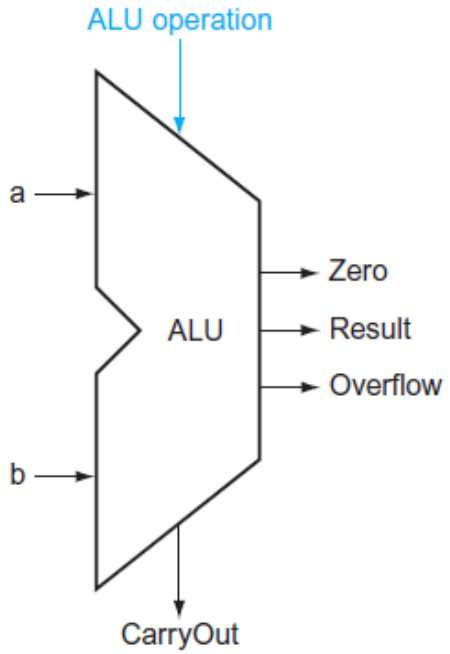


ALU Control

- We can think of the combination of the 1-bit Ainvert line, the 1-bit Bnegate line, and the 2-bit Operation lines as **4-bit control lines for the ALU**, telling it to perform add, subtract, AND, OR, NOR, o set less than.

ALU control lines	Function
0000	AND
0001	OR
0010	add
0110	subtract
0111	set less than
1100	NOR

Arithmetic Logic Unit (ALU)



Symbol for a Complete ALU